

Cloud computing

Cloud computing is the on-demand delivery of computing resources such as servers, storage, databases, networking, and software over the internet on a pay-as-you-go basis.

Why Do We Use Cloud Computing?

- Reduces infrastructure cost.
- Access resources from anywhere.
- Scale resources up or down easily.
- High availability and reliability.
- Faster application deployment.

Characteristics of Cloud Computing

- On-demand self-service – Users can create resources whenever needed.
- Broad network access – Accessible through the internet.
- Resource pooling – Resources are shared among multiple users.
- Rapid elasticity – Resources can be increased or decreased quickly.
- Measured service – Pay only for what you use.

Cloud Service Models

- IaaS (Infrastructure as a Service) – Virtual machines, storage, networking.
- PaaS (Platform as a Service) – Platform to develop and deploy applications.
- SaaS (Software as a Service) – Software accessed through a web browser.

Cloud Deployment Models

- Public Cloud – Shared infrastructure available to everyone.
- Private Cloud – Dedicated to a single organization.
- Hybrid Cloud – Combination of public and private clouds.
- Community Cloud – Shared by organizations with similar requirements.

Advantages

- Cost-effective
- Scalability
- High availability
- Automatic updates
- Disaster recovery
- Global accessibility

Disadvantages

- Internet dependency
- Security and privacy concerns
- Downtime
- Vendor lock-in

Examples of Cloud Providers

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)